Practice workbook · Using a binomial expansion to estimate a value

Choosing a value of x that turns the bracket into the required number and substituting into the truncated series to estimate it.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-20

1. Question 9 · part b · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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1. Question 9 · part b · 3 marks · Working and answer

Working and answer

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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2. Question 9 · part d · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 11 marks)

2. Question 9 · part d · 4 marks · Working and answer

Working and answer

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3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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3. Question 6 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 7 marks)

3. Question 6 · part c · 2 marks · Working and answer

Working and answer

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4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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4. Question 6 · part c · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

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Question 6 continued

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(Total for Question 6 is 8 marks)

4. Question 6 · part c · 3 marks · Working and answer

Working and answer

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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5. Question 6 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part c · 3 marks · Working and answer

Working and answer

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6. Question 1 · part b · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 (a) Expand $\left(1+\frac{x}{4}\right)^8$ in ascending powers of x up to and including the term in x^3

Give each coefficient in its simplest terms.

(3)

(b) Use your expansion with a suitable value of x to obtain an approximation, to 4 decimal places, of $(1.035)^8$

(3)

(Total for Question 1 is 6 marks)

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6. Question 1 · part b · 3 marks · Working and answer

Working and answer

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$ Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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7. Question 9 · part c · 2 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 8 marks)

7. Question 9 · part c · 2 marks · Working and answer

Working and answer

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8. Question 9 · part c.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.i only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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8. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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9. Question 9 · part c.ii · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c.ii only.

Question 9 continued

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9. Question 9 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

Question 9 continued

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9. Question 9 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.



Question 9 continued

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(Total for Question 9 is 9 marks)

9. Question 9 · part c.ii · 4 marks · Working and answer

Working and answer

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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

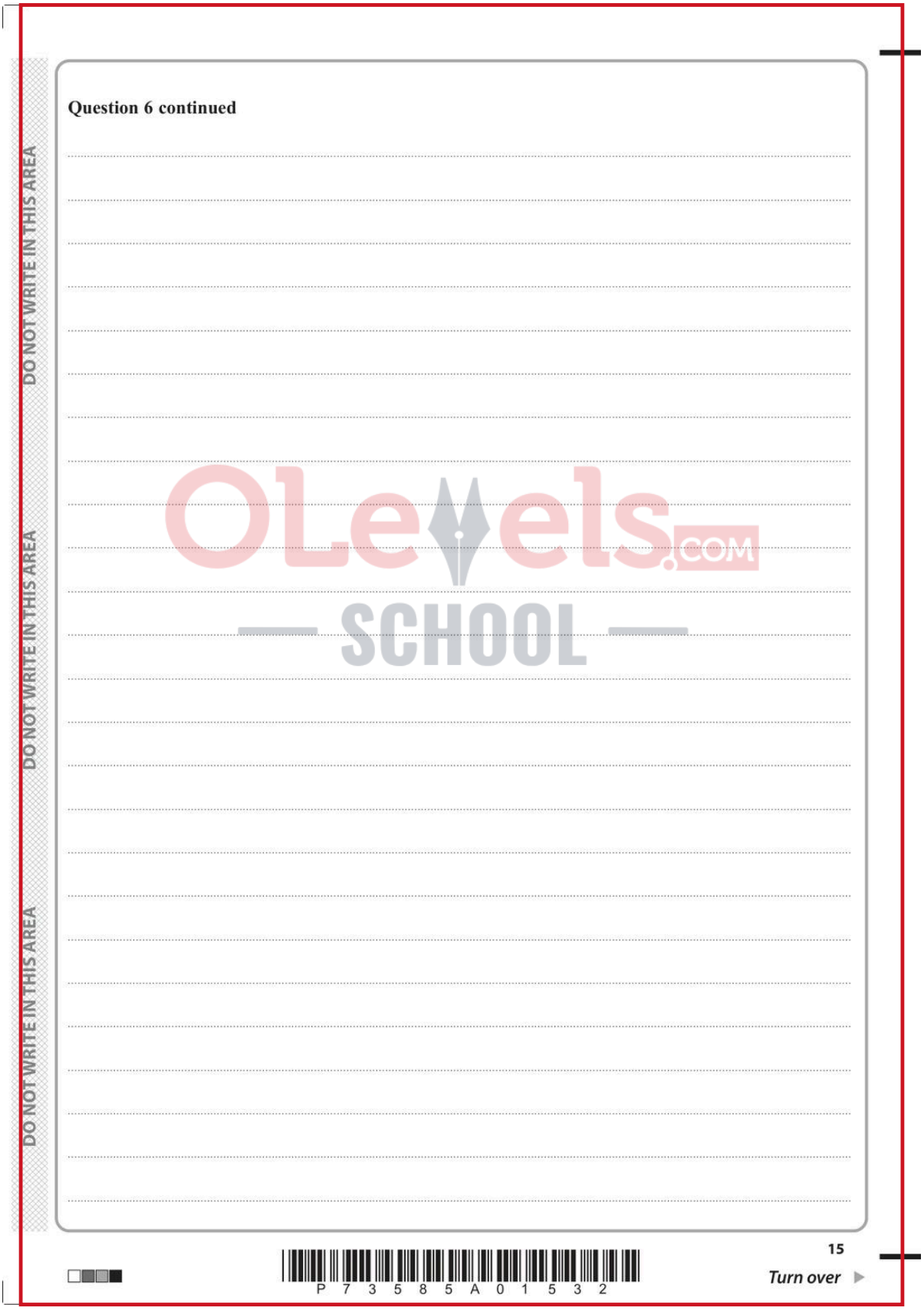
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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.



Question 6 continued

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10. Question 6 · part d · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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10. Question 6 · part d · 3 marks

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Further practice · Attempt part d only.

Question 6 continued

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(Total for Question 6 is 12 marks)

10. Question 6 · part d · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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11. Question 5 · part b · 3 marks · Working and answer

Working and answer

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12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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12. Question 5 · part c · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 5 continued

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(Total for Question 5 is 8 marks)

12. Question 5 · part c · 2 marks · Working and answer

Working and answer

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